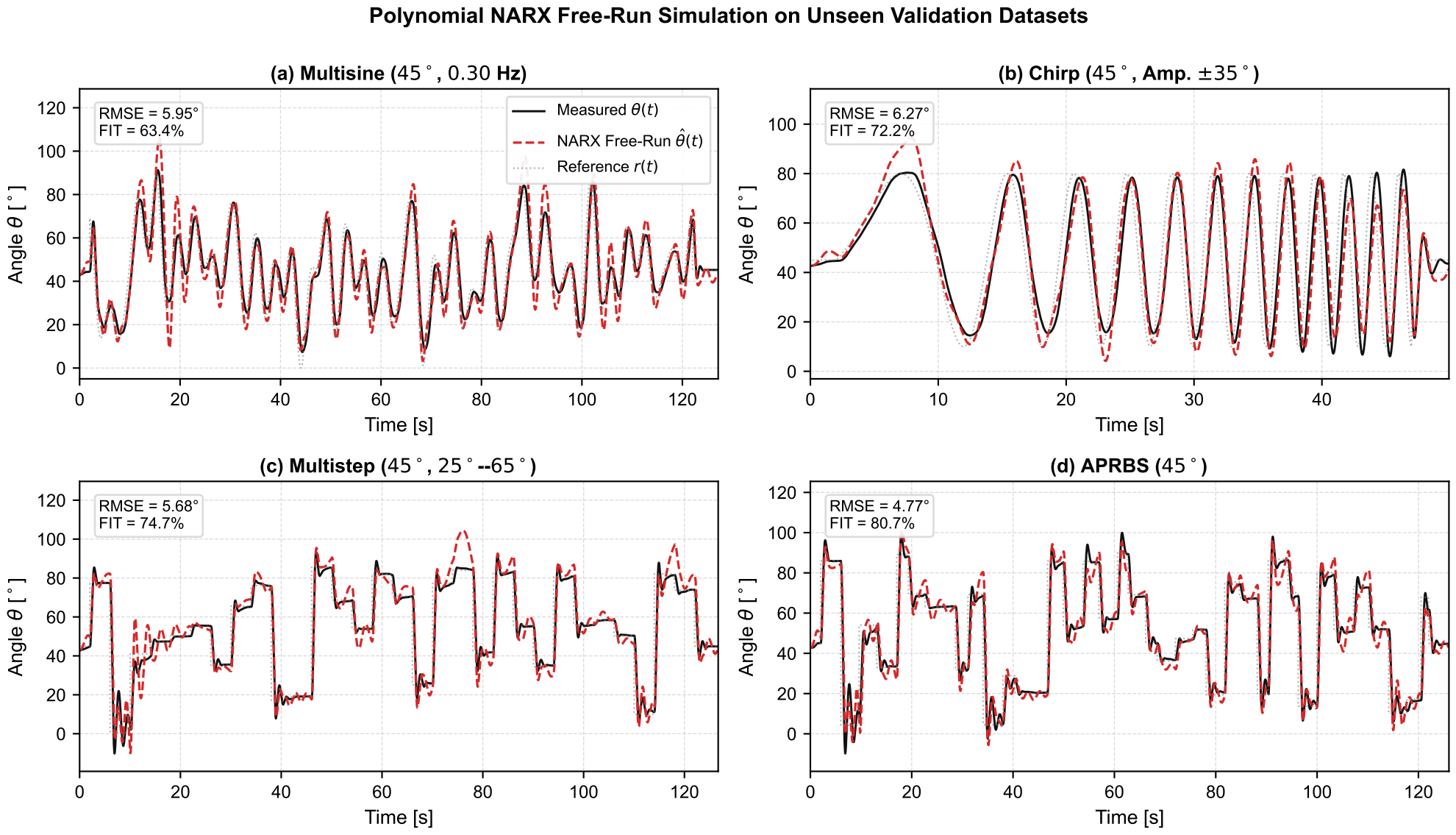

Polynomial NARX Free-Run Simulation on Unseen Validation Datasets
(a) Multisine (45 °, 0.30 Hz)
RMSE = 5.95°
FIT = 63.4%
Measured θ(t)
NARX Free-Run θ̂(t)
Reference r(t)
Angle θ [°]
Time [s]
(b) Chirp (45 °, Amp. ±35 °)
RMSE = 6.27°
FIT = 72.2%
Angle θ [°]
Time [s]
(c) Multistep (45 °, 25 °--65 °)
RMSE = 5.68°
FIT = 74.7%
Angle θ [°]
Time [s]
(d) APRBS (45 °)
RMSE = 4.77°
FIT = 80.7%
Angle θ [°]
Time [s]